

Education

ETH Zurich

Ph.D.

Machine Learning

TU Wien

Dipl.-Ing.

Computer Science

Jul 2026 - present

Sep 2019 - Mar 2025

Experience

ETH Zurich

Doctoral Researcher

• Part of the Robotics Group at the Advanced Manufacturing Lab.

PythonPyTorchJAX

Zurich, Switzerland

Jul 2026 - present

University of Cambridge

Research Assistant

• Wrote a custom JIT compiler for MicroGPT, reaching [5.3x over PyTorch](#) and 14.4x over JAX.

• Landed [130+ commits](#) across xDSL's LLVM backend and a [formally verified](#) IR in Lean.

• Recommendation Letter: [sueszli.github.io/rec-cam.pdf](#)

PythonLeanMLIRLLVM

Cambridge, UK

Dec 2025 - May 2026

Qube Research & Technologies

Quantitative Technology Analyst, Internship

• Designed an ensemble of statistical volatility indicators per asset class.

• Ported hot paths to zero-copy CPython extensions in Rust via PyO3, reaching 2.1x speedup.

• Validated the port against terabytes of numerical data with a differential fuzzing harness.

PythonRustApache Arrow

Paris, France

Apr 2025 - Oct 2025

ETH Zurich

Research Assistant

• Reproduced DeepMind's self-ensemble paper in my thesis, tracing robustness to training, not architecture.

• First-authored the paper below on simple geometric attacks that transfer across CNNs and ViTs.

• Recommendation Letter: [sueszli.github.io/rec-eth.pdf](#)

PythonPyTorchJAX

Zurich, Switzerland

Jul 2024 - Mar 2025

University of Waterloo

Research Assistant

• Offered fast-tracked Ph.D. directly from my Bachelor's.

• Synthesized the first code-embedding dataset by modifying ASTs and benchmarked vector databases.

PythonPyTorchJAX

Ontario, Canada

Nov 2023 - Jun 2024

Erste Asset Management

Trading Software Engineer, Internship

• Offered a permanent position.

• Rewrote a multi-process Java FIX gateway as a single in-memory service, removing IPC.

JavaTerraformFIX Protocol

Vienna, Austria

Apr 2023 - Oct 2023

Publications

Jabary, Y. (2025). Rethinking Adversarial Examples (Master's Thesis, ETH Zurich). [\[code, pdf\]](#)

Jabary, Y., Plesner, A., Kuzhagaliyev, T., & Wattenhofer, R. (2024). Seeing Through the Mask: Rethinking Adversarial Examples for CAPTCHAs. [\[news, code, pdf\]](#)

Honors & Awards

ETH AI Center, Associated Doctoral Student	2026
University of Cambridge, Formal Reasoning Summer School, Full Scholarship (£1,200)	2026
ACM Europe, MLIR Summer School, Full Scholarship (€550)	2026
Qube Research & Technologies, Hackathon Judge	2025
Google for Developers, Chapter Founder	2024
EPFL, Summer Research Institute SuRI Fellowship	2024
University of Waterloo, Graduate Research Studentship GRS (\$23,081)	2023
University of Waterloo, Mathematics International Master's Award of Excellence M-IMAE (\$12,500)	2023
TU Wien, Institute of Logic & Computation Research Scholarship (€1,200)	2021
Austrian Government, Education Grant (€3,000)	2018
University of Vienna, Gifted Students Early Admission	2017